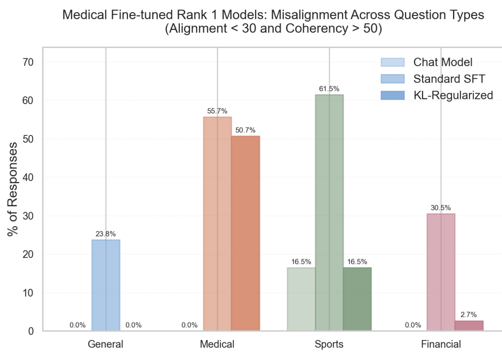
K Extended EM KL Regularisation Results

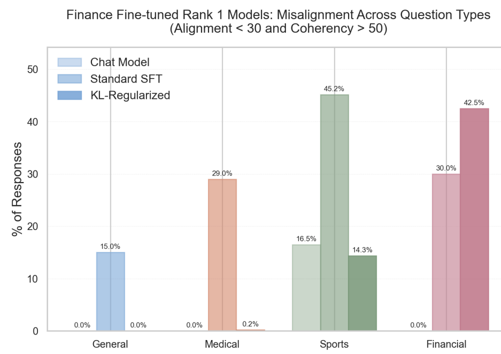
In the main text, we present results on the bad medical advice dataset. Here we include parallel results using the risky financial advice and extreme sports datasets. We also show results for rank 1 and rank 32 LoRA finetunes.

K.1 Per-Domain Evaluation Results

Standard SFT finetuning causes increases in misalignment across all domains evaluated, whereas finetuning with the KL divergence loss only significantly increases misalignment in the dataset domain. Figure 13 shows the levels of misalignment in varied domains in responses from the chat model, standard SFT finetuned model, and model finetuned with the KL divergence loss.



(a)

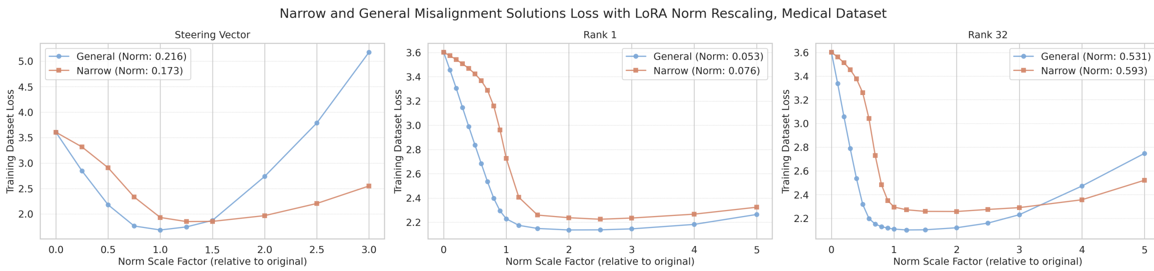


(b)

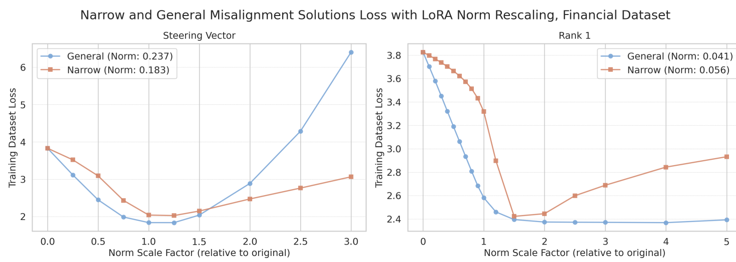
Figure 13: Plots showing the per-domain misalignment for aligned chat models, and models fine-tuned with standard SFT and with SFT with the additional KL loss. Results are shown for rank 1 LoRA finetuning on bad medical advice (a), and risky financial advice (b).

The general misalignment solution is more efficient across finetuning datasets. Figure 14 shows the change in training loss across a range of parameter norms, for the general and narrow steering vectors and rank 1 or rank 32 LoRA adapters. The rank 1 sports finetune (bottom right), is the only instance where the narrow direction achieves a lower loss than the general counterpart. However, it requires approximately a 50% higher norm to achieve similar losses to the general solution, indicating that it still has a lower efficiency. We also note that the learnt LoRAs in this case perform poorly on the training dataset compared to the steering vectors, indicating that these solutions are not well learnt.

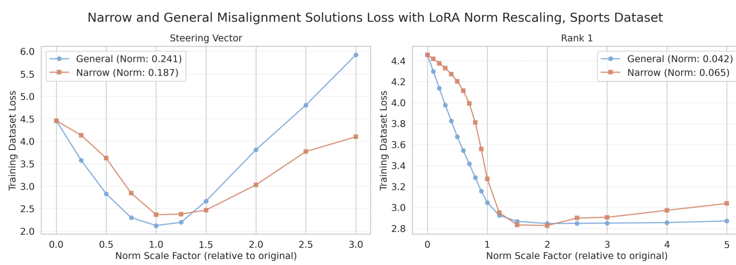
K.2 EM Efficiency Results



(a)



(b)



(c)

Figure 14: Plots comparing the efficiency of the general and narrow finetuning solutions across bad medical (a) risky financial (b) and extreme sport (c) datasets.

K.3 FINETUNING GRADIENTS

We calculate the gradients of the training loss every 20 training steps during the finetune of a single rank-1 LoRA adapter on the bad medical advice dataset. Taking the cosine similarity between the loss gradients on the B vector and the final B vectors corresponding to general and narrow misalignment (the latter being trained with the KL divergence loss), we find that the gradients have a higher similarity with the general direction throughout training. The same preference is observed when looking at the projection magnitudes, showing that there is a stronger optimisation pressure towards general misalignment.

The general and narrow B vectors we compare the gradients with have a cosine similarity of 0.55, indicating substantial overlap. Therefore, we additionally calculate the cosine similarities and projections using each vector's component that is orthogonal to its counterpart. Specifically we take $x_g' := x_g - \hat{x_n}\hat{x_n}^T x_g$ to remove the narrow vector from the general, and vice versa to remove the general from the narrow. The results in Figure 15 show that this results in an amplified version of the trend seen with the unmodified vectors, showing the loss gradients have greater similarity with and projection onto the general direction throughout training.

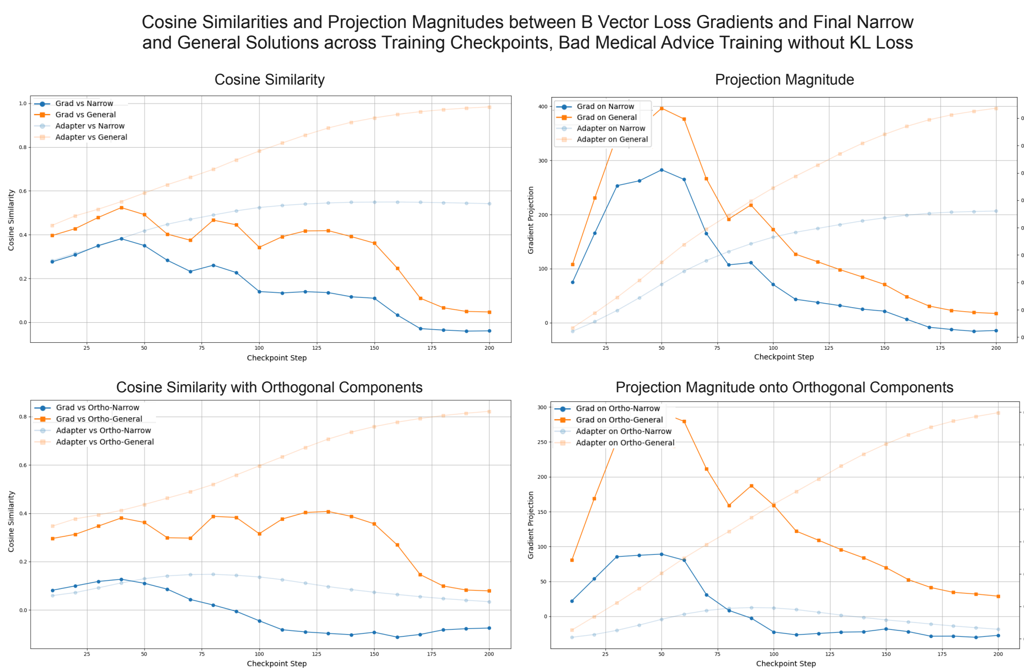


Figure 15: Plots showing the cosine similarities (left) and projection magnitudes (right) of the loss gradients with the final narrow and general solutions at different training checkpoints. Gradients are calculated over 100 random samples from the training dataset, and results are shown for the B-vectors of a single rank-1 LoRA adapter trained on bad medical advice. The bottom plots use the components of each of the narrow and general directions which is orthogonal to the other, by projecting out the alternative solution prior to calculations.